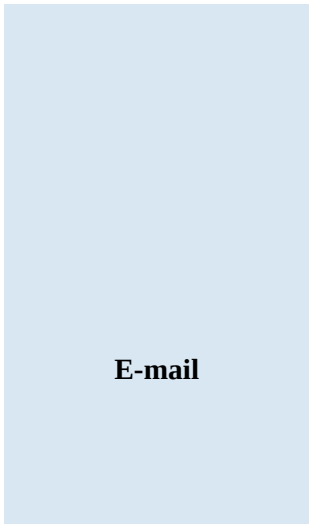


# 一种博弈驱动的多智能体语义共识方法及影视推荐系统


**E-mail**


2026 10

# 0 缩略语和关键术语定义

| 术语 | 定义 |
| --- | --- |
| AEIC | Assumption Evidence Inference Conclusion |
| | Agent |
| | Agent AEIC / |
| / | |
| | Agent |
| | Agent P2P |
| | Agent |
| | |

# 1 相关技术背景及最接近的现有实现方案

## 1.1

Agent

Agent Agent “ ” “ ”

Agent Top-K Agent

## 1.2

### 1.2.1 现有实现方式

Agent

### 1.2.2 现有方案的不足

1 Agent

2

3

4

## 1.3 Agent

### 1.3.1 现有实现方式

Agent

### 1.3.2 现有方案的不足

1 Agent

2

3 Agent Top-K

4

## 1.4

### 1.4.1 现有实现方式

### 1.4.2 现有方案的不足

1 Top-K

2

3

4

# 2 本发明技术方案的详细阐述

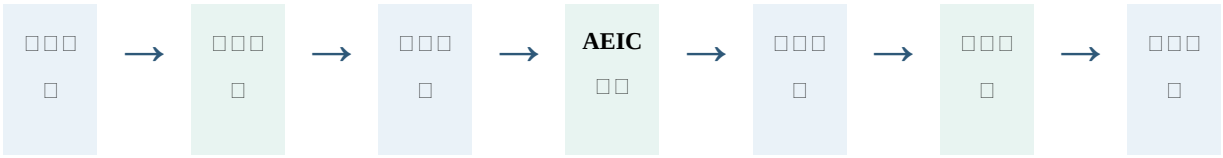
## 2.1

Agent Agent

Agent

Top-K

## 2.2




1

1 “ ” Trace Agent Token “ ”

### 2.2.1 □□□□□□□□□□□□

q Agent N={1,…,n} t={1,…,T} S_t Agent Agent

$$S\_t = (q,\ m,\ t,\ \rho(t-1),\ P\_t,\ B\_t,\ H\_t)$$

ρ(t−1) t P_t B_t H_t Agent i view_i(S_t) Agent Agent

τ_c τ_a τ_r τ_c<τ_a<τ_r Agent Agent

### 2.2.2 □□ S1□□□□□□□□□

m g c P B Agent

$$m = (g,\ c,\ P,\ B)$$

Agent Token Agent

JSON g c P Agent B Token

| □□□□ | □□ | □□□□ |
|---|---|---|
| g | | |
| c | Low AEIC Verify Imitate | |
| P | Agent× × | Agent |
| B | Token | |

### 2.2.3 步骤 S2：Agent 独立选择行动

Agent A Low AEIC Verify Imitate

| 行动 | 真实执行路 □ | 可记录资源 |
|---|---|---|
| Low | Token | / Token |
| AEIC | | Token |
| Verify | AEIC | |
| Imitate | | / |

Agent action reason_code expected_credit expected_cost reason_code expected_credit expected_cost Agent action Low

$$a\_i(t) \in \{Low, AEIC, Verify, Imitate\}$$

### 2.2.4 步骤 S3：按行动路由实际资源

Agent Verify Low Token

DeepSeek API Agent “ ” “ ”

action executor Low AEIC Verify AEIC Imitate P Agent action_trace

$$R\_i = route(a\_i, P\_i, B\_i)$$

R_i Agent i a_i R_i Verify Imitate B_i Agent

### 2.2.5 步骤 S4：生成 AEIC 结构化语义报告

AEIC Assumption Evidence Inference Conclusion / /

| 字段 | 主要内容 | 可计算信号 |
|---|---|---|
| A | | field_A |

| 字段 | 主要内容 | 可计算信号 |
| --- | --- | --- |
| E | | field_E novel_evidence verified_evidence |
| I | / | field_I consistency useful_correction |
| C | | field_C unsupported_flip |

AEIC “ ” “ ”

Evidence evidence_id claim source_type source_ref status span source_type source_ref status span verified_evidence

Conclusion / / support oppose abstain preference_score risk_flags Trace

### 2.2.6 步骤 S5：提取可观察语义信号

s_i

A E I C Agent

$$completeAEIC\_i = 1[A\_i \neq \varnothing \wedge E\_i \neq \varnothing \wedge I\_i \neq \varnothing \wedge C\_i \in Y]$$

$$novelEvidence\_i = |tokens(E\_i) \setminus tokens(E\_visible, i)| / max(1, |tokens(E\_i)|)$$

useful_correction Agent unsupported_flip

$$unsupportedFlip\_i(t) = 1[C\_i(t) \neq C\_i(t-1) \wedge \Delta E\_i(t) = 0 \wedge verify\_i(t) = 0]$$

verified_evidence

unknown

### 2.2.7 步骤 S6：计算信用、成本和已实现效用

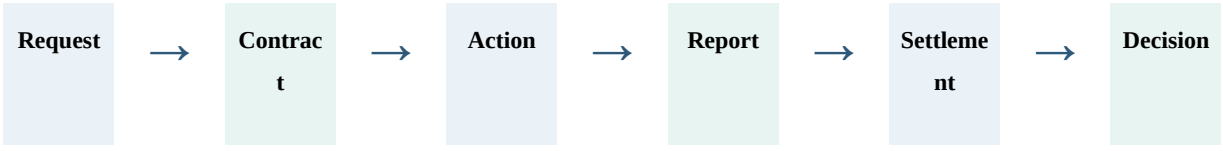
AEIC

$$credit\_i = \Sigma\_k \; w\_k \cdot s\_k(r\_i, h\_i, y)$$

$$utility\_i = credit\_i - cost\_i(action\_i)$$

r_i Agent i h_i Agent y

AEIC



$$credit\_i = clip(w\_c \cdot complete\_i + w\_n \cdot novel\_i + w\_v \cdot verified\_i + w\_u \cdot correction\_i - w\_f \cdot fli$$

y outcome

### 2.2.8 步骤 S7：批量更新信誉

$$reputation\_i(t) = decay \cdot reputation\_i(t-1) + utility\_i(t)$$

j reputation_before Agent u_ij

reputation_after

$$\rho\_i(after) = decay \cdot \rho\_i(before) + mean\_j(u\_ij)$$

agent_id

### 2.2.9 步骤 S8：形成可审计集体决策

Agent

y Agent ω_i Agent

support oppose

$$\omega\_i = clip(conf\_i \cdot (1 + \rho\_i / K) \cdot max(0, utility\_i),\ \omega\_min,\ \omega\_max)$$

$$consensus(y) = \Sigma\_i\ \omega\_i \cdot 1[C\_i = y] - \lambda \Sigma\_i\ \omega\_i \cdot 1[C\_i\ opposes\ y]$$

Agent

“ ”

### 2.2.10 数据结构和审计 Trace

2

request_id Trace AEIC

| 记录对象 | 必需字段示例 |
| --- | --- |
| | request_id |
| | contract_version g c P B committed_at |
| | agent_id action visible_state selected_at |

| 记录对 □ | 必需字段示例 |
| --- | --- |
| | A/E/I/C evidence_refs raw_response tokens latency |
| | signals credit_breakdown cost utility reputation_before/after |
| | normalized_result support/oppose residual_dissent rank score_components |

Trace event_id request_id event_type actor timestamp payload_hash prev_hash

Trace

$$hash\_e = H(request\_id \;||\; event\_type \;||\; payload\_hash \;||\; prev\_hash\_e)$$

Agent× ×

complete incomplete

## 2.3

Agent Trace

Agent

| 模块 | 输入 | 输出 | 关键约束 |
| --- | --- | --- | --- |
| | Top-K | | |
| | | | ID |
| | | | Agent |
| Agent | | AEIC | |
| | | / | |
| | | | |
| Trace | | | |

### 2.3.1 请求解析与风险预算规划

/

Top-K

Agent

query required_genres preferred_genres excluded_genres min_year max_year top_k “ ” excluded_genres “ ” preferred_genres

Agent

$$risk = clip(b\_0 + b\_h \cdot n\_hard + b\_n \cdot n\_negative + b\_q \cdot complexity + b\_k \cdot topK, 0, 1)$$

### 2.3.2 真实目录召回与冻结候选集

18,430 MovieLens

Top-K Agent candidate_id

Top-K

ID

movie_id title year genres retrieval_score catalog_evidence

$$C\_frozen = topL(\{x \in Catalog \mid hardPass(x,q)=1\}, retrievalScore)$$

L top_k candidate_id L Agent C_frozen

### 2.3.3 多角色 Agent 对统一候选集进行 AEIC 评审

Agent Agent Agent Agent Agent AEIC Agent

j z_j z_ij=1 Agent i z_ij=0 N_req j coverage_incomplete Agent

$$coverage(j) = \Sigma\_\{i \in N\_req\} z\_ij / |N\_req|$$

### 2.3.4 硬约束门控与候选级共识排序

Agent

“ Top-K ”

$$score\_j = \alpha \cdot retrieval\_j + \beta \cdot agentUtility\_j + \gamma \cdot evidence\_j - \mu \cdot opposition\_j$$

opposition_j Top-K

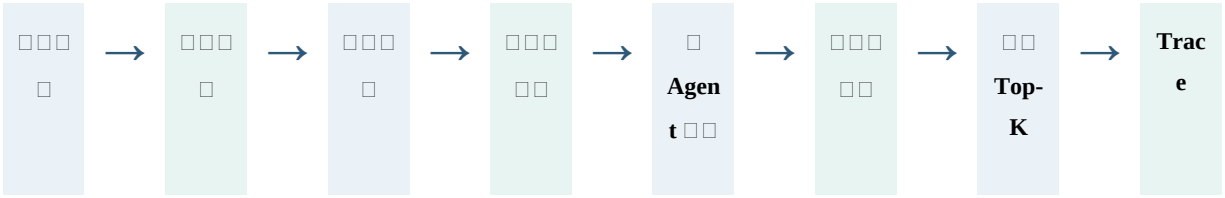
hardPass_j Agent

$$hardPass\_j = yearPass\_j \wedge requiredGenrePass\_j \wedge excludedGenrePass\_j$$

$$TopK = argsort\_K(\{score\_j \mid j \in C\_frozen \wedge hardPass\_j = 1 \wedge coverage(j) = 1\})$$

K score_components

Agent




3

### 2.3.5 □□□□□□□□□□□□□□□□

“ ” Agent AEIC Trace

Token A/E/I/C

request_id Trace

### 2.3.6 □□□□□□□□□□□□

HTTP

request_id

Agent

mechanism_status degraded “ Agent ”

| □□ | □□ | **Trace □□** |
|---|---|---|
| | Low | invalid_action fallback |
| AEIC | | schema_repair raw_hash |
| Agent | | timeout retry coverage |
| | Top-K | verification_unavailable |
| | | degraded missing_agents |
| | request_id | client_disconnected |

## 2.4

### 2.4.1 □□□□□□□□□□□□

AEIC Verify Low Imitate

Agent Agent

A E I C P

Agent Agent

Agent Verify unsupported_flip

useful_correction verified_evidence

| 阶段 | 控制器动作 | **Agent** 或工具输出 | 审计重点 |
|---|---|---|---|
| | g c P B | | |
| | | action | |
| | | Token | |
| | | A/E/I/C | |
| | | | |
| | | | |
| | | | |

### 2.4.2 多约束影视请求实施例

“ 2010 ” Top-K 3 min_year=2010 required_genres={Science Fiction, Drama} excluded_genres={Action, Disaster} “ ”

18,430 L movie_id

AEIC

Science Fiction Drama Action hardPass Top-K evidence

“ ” A/E/I/C

### 2.4.3 关键算法流程

1 q request_id

2 m committed

3 Agent view_i(S_t) a_i route(a_i)

4 AEIC s_i

5 g credit_i c(a_i) utility_i

6

7 reputation_after

8 complete incomplete

## 2.5

Agent StrategyQA MMLU-Hard TruthfulQA 150 450 5 Agent 3 Seed

| 比较条件 | 高资源行动率 | 第三轮多数决策准确率 |
| --- | --- | --- |
| No Mechanism | 0.46% | 68.00% |
| AEIC Static AEIC | 56.93% | 82.22% |
| | +56.47 | +14.22 |

Exact McNemar Holm $p=1.23\times10^{-11}$ “ ”

30 150 Top-5 Agent

StrategyQA 73.33% 77.33% MMLU-Hard 71.33% 91.33% TruthfulQA 59.33% 78.00%

3 5 Agent AEIC

## 2.6

1 Agent

2 AEIC

3 Verify AEIC Token

4 –

5

6

7 Trace

## 2.7

1 Agent

2

3

4 Agent

5 Agent

6 Agent Trace Agent

7 AEIC JSON

8

9

## 3 本发明的关键技术点和拟保护点

1.

2. — Agent Token

3. AEIC

4. —

5.

6. Agent

7. Agent Top-K

8. request_id AEIC

## 4 附件及参考资料

| 材料 | 位置/说明 |
| --- | --- |
| | docs/papers/ experiments/results/ AEIC |
| | system/game_consensus_recsys/ Agent HTTP Trace |
| | experiments/results/movie_acceptance/ acceptance_results.jsonl 30 |
| | docs/patent/source/ .docx |
| | docs/patent/source/ .docx |

“ ”“ ”“ ” “

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